



New electrolytes using Li_2O or Li_2O_2 oxides and tris(pentafluorophenyl) borane as boron based anion receptor for lithium batteries

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ABSTRACT

A new system of electrolytes has been developed and studied for lithium-ion batteries. This new system is based on the interactions between Li_2O or Li_2O_2 and tris(pentafluorophenyl) borane (TPFPB) in carbonate based organic solvents. This opens up a completely new approach in developing non-aqueous electrolytes. In general, the solubility of Li_2O or Li_2O_2 is very low in organic solvents and the ionic conductivities of these solutions are almost undetectable. By adding certain amount of tris(pentafluorophenyl) borane (TPFPB), one type of boron based anion receptors (BBARs), the solubility of Li_2O or Li_2O_2 in carbonate based solvents was significantly enhanced. In addition, the Li^+ transference numbers of these new electrolytes measured were as high as 0.7, which are more than 100% higher than the values for the conventional electrolytes for lithium-ion batteries. The room-temperature conductivities are around 1×10^{-3} S/cm. These new electrolytes are compatible with LiMn_2O_4 cathode for lithium-ion batteries.

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1. Introduction

There is now a search for new non- LiPF_6 electrolytes with high lithium ion transference number, larger electrochemical stability window, wide operating temperature and low sensitivity to moisture, aiming to extend the application lithium-ion batteries into electric vehicles (EVs), hybrid electric vehicles (HEVs), and plug-in hybrid electric vehicles (PHEVs) [1]. One innovative approach is to use boron based anion receptors (BBARs) to complex the anion of the lithium salt (LiX), forming a LiX -BBAR series of electrolytes. Various BBAR compounds have been synthesized and studied at BNL over last decade [2–5]. Due to the strong interaction between the anion in LiX and BBAR, it was found that the insoluble LiF can be dissolved in carbonate or ether based solvents to form novel electrolytes. A typical LiX -BBAR electrolyte is formed by using tris(pentafluorophenyl) borane (TPFPB) as an anion acceptor and LiF as the lithium salt and propylene carbonate (PC), ethyl carbonate (EC) or dimethyl carbonate (DMC) as solvents. It was observed recently that the Li^+ transference numbers of these electrolytes are as high as 0.6 and the room-temperature conductivity can reach in a range of 2×10^{-3} S/cm [6]. Therefore, the effective lithium ion conductivity is quite high, which is defined as the product of the transference number and the conductivity. It is this lithium ion effective conductivity, not just the total conductivity, playing the

key role for achieving small polarization in battery operation. In addition, it was also found that the electrolytes containing propylene carbonate (PC) show quite good low-temperature conductivity. Inspired by these results, it will be quite interesting to see, if the insoluble Li_2O or Li_2O_2 can also be dissolved in carbonate based solvents by reacting with the boron based anion receptor and form new electrolytes with high effective conductivities.

In this communication, the results of our studies on the transport properties and electrochemical performance of these new electrolytes using TPFPB- Li_2O / Li_2O_2 in carbonate solvents in lithium batteries are reported.

2. Experimental

Tris(pentafluorophenyl) borane (TPFPB) was synthesized in Brookhaven National Laboratory as described in previous work [2]. Li_2O and Li_2O_2 were purchased from Aldrich. Both TPFPB and lithium oxides were dried under vacuum at 80 °C for 12 h before use. Ethylene carbonate (EC), propylene carbonate (PC) and dimethyl carbonate (DMC) (All from Shanghai Topsol, Battery grade, $\text{H}_2\text{O} < 5$ ppm) were dried through 4 Å size molecular sieves in an argon-filled glove box. Three electrolytes with nominal molar concentrations were prepared: 0.4 M TPFPB and 0.2 M Li_2O in PC/DMC (1:1 in volume) solution (named as TPLO-PC-01); 0.4 M TPFPB and 0.2 M Li_2O_2 in PC/DMC (1:1 volume) solution (named as TPLOO-PC-01); 0.4 M TPFPB and 0.2 M Li_2O in EC/DMC (1:1 volume) solution (named as TPLO-EC-01). The other three electrolytes (0.2 M

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Li₂O in PC/DMC (1:1), 0.2 M Li₂O₂ in PC–DMC (1:1), 0.4 M TPFPB in PC/DMC (1:1)) were also prepared for comparison. Conductivity measurements were performed using a Novacontrol beta high performance impedance analyzer in the frequency range from 1 Hz to 1 MHz. A standard glass cell with two Pt electrodes was used for the conductivity measurements. The cell constant was determined using a standard 0.01 M KCl aqueous solution for every measurement. The lithium ion transference number was obtained by combining the ac impedance and dc polarization measurement using the same Li/electrolyte/Li cell [6].

LiMn₂O₄ (Wuxi Jewel, China) was obtained as commercial products and was used as it is. The electrodes for battery testing were prepared using 90 wt% active material, 5 wt% acetylene black and 5 wt% poly(vinylidene fluoride) (PVDF). Aluminum foil was used as current collectors for cathode. The testing battery was a Swagelok-type two-electrode cell using a lithium foil as anode and a Celgard 2300 as separator. Charge-discharge test was carried out on a Land battery test system. The conventional LiPF₆ based electrolyte (1 M LiPF₆ in EC/DMC 1:1) used as reference was obtained as a commercial product from Shanghai Topsol. The battery cycling measurements were conducted at room temperature.

3. Results and discussion

Without TPFPB, both Li₂O and Li₂O₂ show poor solubility in the PC–DMC (1:1 in v/v) solvent (Measured solubility of Li₂O and Li₂O₂ in both cases are 0.01 mol/L and 0.02 mol/L, respectively). The conductivities of both the 0.2 M Li₂O in PC–DMC (1:1 in v/v) and 0.2 M Li₂O₂ in PC–DMC (1:1 in v/v) solution were very low, as shown in Fig. 1. It was observed that the solubility of Li₂O₂ in PC–DMC is higher than that of Li₂O indicated also by the conductivity of

0.2 M Li₂O₂ in PC–DMC electrolyte is one order of magnitude higher than that of 0.2 M Li₂O in PC–DMC. This is perhaps related to the weaker Li–O bonding in Li₂O₂ than Li₂O (the bond length of Li–O in Li₂O₂ is 2.15 Å [7] vs the 1.996 Å bond length of Li–O in Li₂O [8]).

After adding 0.4 M TPFPB, the solubility of Li₂O or Li₂O₂ in PC–DMC is enhanced to 0.19 mol/L for both cases. It can be seen in Fig. 1 that the conductivity of the TPLO–PC-01 electrolyte shows three orders of magnitude higher than that of 0.2 M Li₂O in PC–DMC (1:1). This also indicates clearly that the adding of TPFPB promotes the solubility of Li₂O significantly. Similar situation can be observed in the case of Li₂O₂.

It is also noticed that the PC-based electrolytes show much higher conductivity than EC-based electrolyte in low temperature range (below 0 °C). This might be due to the low melting point of PC (~–49 °C) than EC (~–36 °C) [1]. The activation energies for different electrolytes in two temperature ranges are listed in Table 1. Conductivity values of different electrolytes at several temperatures are listed in Table 2. It can be seen that these three LiX–TPFPB electrolytes have moderate conductivities at room temperature.

The Li ion transference number were measured by combining AC impedance and DC polarization on the same cell containing the same sample using two lithium foils as non-blocking electrodes [6]. The results are listed in Table 3. It can be seen that the Li⁺ transference number for every electrolyte is at least 100% higher than that of 1 M LiPF₆ (0.21) or LiClO₄ (0.34) or LiBF₄ (0.29) in PC–DMC (1:1) electrolytes [6]. It is supposed that B^{δ+} center in TPFPB interacts with O in Li₂O strongly and form large coordinate anions with low mobility and free more Li⁺ at the same time, resulting in the high transference number of Li⁺. More spectroscopic studies and theoretical calculations are under way to investigate this issue further.

The higher Li ion transference number can achieve higher effective Li ion conductivity (defined as total conductivity multiplies the Li ion transference number). Therefore, although the 1.4×10^{-3} S/cm total conductivity for TPLO–PC-01 is not impressive comparing with the 8.8×10^{-3} S/cm value for 1.0 M LiPF₆ in EC–DMC, the 1.1×10^{-3} S/cm effective Li-ion conductivity of TPLO–PC-01 (1.4×10^{-3} S/cm, $t_{\text{Li}^+} = 0.78$) at 20 °C is quite close to the 1.8×10^{-3} S/cm value of 1.0 M LiPF₆ in EC–DMC (8.8×10^{-3} S/cm, $t_{\text{Li}^+} = 0.21$) at 20 °C.

Table 1

E_a comparison of different electrolytes in high and low temperature ranges

No.	Electrolyte	<i>E_a</i> (kJ/mol)	
		High temperature	Low temperature
a	0.4 M TPFPB–0.2 M Li ₂ O–PC:DMC (1:1)	6.0	14.1
b	0.2 M Li ₂ O–PC:DMC (1:1)	5.5	12.6
c	0.4 M TPFPB–PC:DMC (1:1)	8.0	8.0
d	0.2 M Li ₂ O ₂ –PC:DMC (1:1)	6.0	16.8
e	0.4 M TPFPB–0.2 M Li ₂ O ₂ –PC:DMC (1:1)	5.9	13.7
f	0.4 M TPFPB–0.2 M Li ₂ O–EC:DMC (1:1)	7.3	65.5

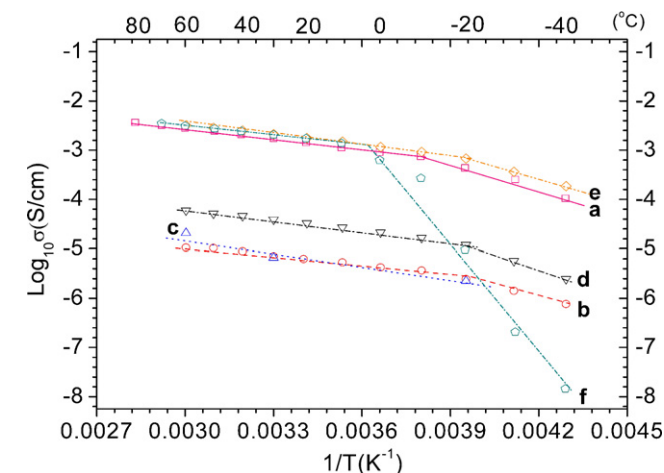


Fig. 1. Temperature-dependent conductivities of different electrolytes. a: TPFPB–Li₂O–PC–DMC; b: Li₂O–PC–DMC; c: TPFPB–PC–DMC; d: Li₂O₂–PC–DMC; e: TPFPB–Li₂O₂–PC–DMC; f: TPFPB–Li₂O–EC–DMC.

Table 2

Conductivity comparison of different electrolytes at several temperatures

No.	Electrolyte	Conductivity (mS/cm)				
		–40 (°C)	–20 (°C)	0 (°C)	20 (°C)	60 (°C)
a	0.4 M TPFPB–0.2 M Li ₂ O–PC:DMC (1:1)	0.105	0.434	0.901	1.43	2.82
b	0.2 M Li ₂ O–PC:DMC (1:1)	7.55e–4	0.00246	0.00424	0.00620	0.0105
c	0.4 M TPFPB–PC:DMC (1:1)	–	0.00225	–	–	0.0207
d	0.2 M Li ₂ O ₂ –PC:DMC (1:1)	0.00246	0.0119	0.0214	0.0329	0.0594
e	0.4 M TPFPB–0.2 M Li ₂ O ₂ –PC:DMC (1:1)	0.186	0.673	1.16	1.76	3.18
f	0.4 M TPFPB–0.2 M Li ₂ O–EC:DMC (1:1)	2.26E–5	0.00952	0.624	1.72	3.10

Table 3
Transfer number measurement on electrolytes at room temperature

No.	Electrolyte	t_{Li^+}	t_{X^-}
a	0.4 M TFPFB–0.2 M Li_2O –PC:DMC (1:1)	0.78	0.22
e	0.4 M TFPFB–0.2 M Li_2O_2 –PC:DMC (1:1)	0.74	0.26
f	0.4 M TFPFB–0.2 M Li_2O –EC:DMC (1:1)	0.89	0.11

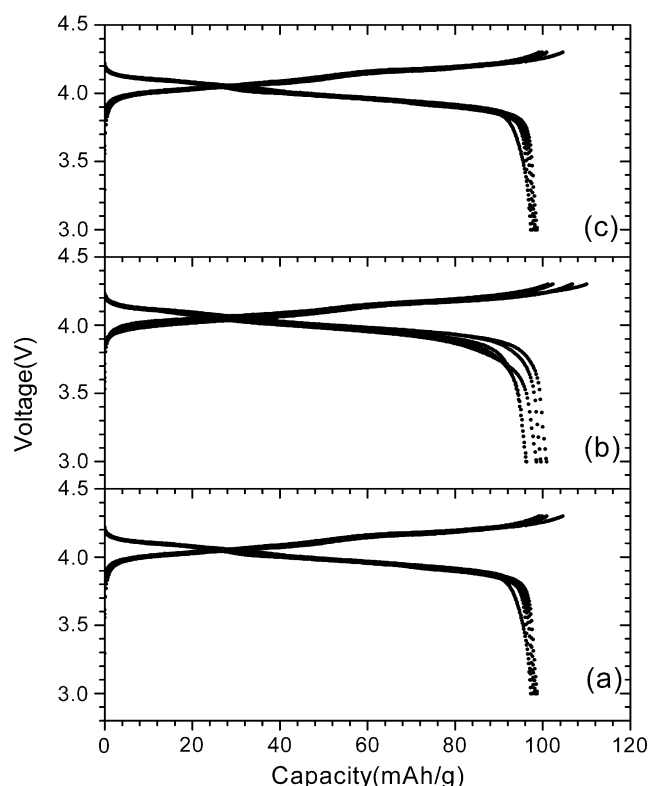


Fig. 2. Charge-discharge curves for Li/LiMn₂O₄ cells at room temperature, with various electrolytes: (a) 0.4 M TFPFB and 0.2 M Li_2O in PC/DMC (1:1, v/v); (b) 0.4 M TFPFB and 0.2 M Li_2O_2 in PC/DMC (1:1, v/v); (c) conventional LiPF_6 based electrolyte as the electrolytes. The cells were cycled between 3 V and 4.3 V at 0.2 C.

The charge-discharge curves of a Li/LiMn₂O₄ half cell using TPLO–PC-01, TPLLO–PC-01 and the commercial LiPF_6 based electrolytes are shown in Fig. 2. The LiMn_2O_4 used in this study is a surface coated material [9]. As shown in Fig. 2, the initial Coulomb efficiencies are 92% and 89%, and the reversible capacities are

98 mAh/g and 101 mAh/g for TPLO–PC-01 and TPLLO–PC-01, respectively. This is similar to the Li/LiMn₂O₄ half cell using conventional LiPF_6 based electrolyte (See curve (c)), and much better than the LiF –TFPFB-based electrolyte under similar conditions [6]. The improved initial Coulombic efficiency in the case of Li_2O –TFPFB electrolyte could be related to its unique surface reaction properties. The compatibility of this new electrolyte with other cathode and anode materials will be reported later in a full paper.

4. Conclusion

It is demonstrated that Li_2O and Li_2O_2 can be dissolved in carbonate based solvents when certain types of boron based anion receptors, such as TFPFB, is added. These new electrolytes show moderate conductivity at room temperature. When PC was used in this type of new electrolytes, the conductivity at low temperatures is quite good. The Li ion transference numbers of these new electrolytes are much higher than those for conventional electrolytes, resulting in high effective lithium ion conductivities. These Li_2O –electrolytes show good compatibility with LiMn_2O_4 cathode materials, showing a potential application in Li ion batteries.

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